AP PRECALCULUS

Unit 1 • Lesson 1.1 • Notes (filled in)

Topic 1.1 • Change in Tandem

Notes (filled in)

Learning goals

- Describe how output values change as input values increase.
- Identify intervals where a function is increasing or decreasing, and where its graph is concave up or concave down.
- Identify zeros and the sign of a function from a graph, a table or a formula.

Vocabulary. function, input, output, increasing, decreasing, rate of change, concave up, concave down, zero, x-intercept, positive, negative.

1. What is a function?

A **function** is a rule that assigns to each input value exactly one output value. We write $y = f(x)$: x is the input (independent variable) and $f(x)$ is the output (dependent variable). As the input values change, the output values change *in tandem*.

2. Increasing and decreasing

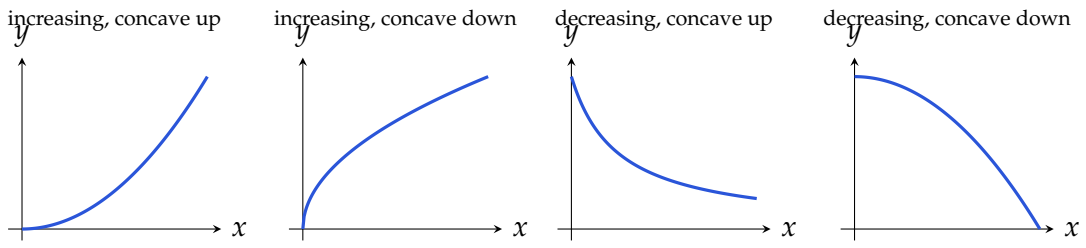
Key idea

On an interval, f is **increasing** if, as the input values increase, the output values always increase: if $a < b$ then $f(a) < f(b)$.
f is **decreasing** if, as the input values increase, the output values always decrease: if $a < b$ then $f(a) > f(b)$.
Intervals of increase and decrease are written with input (x) values.

3. Concavity: how is the function changing?

Key idea

The graph of f is **concave up** on an interval if the rate of change of f is increasing: the graph bends upward.
The graph is **concave down** if the rate of change is decreasing: the graph bends downward.

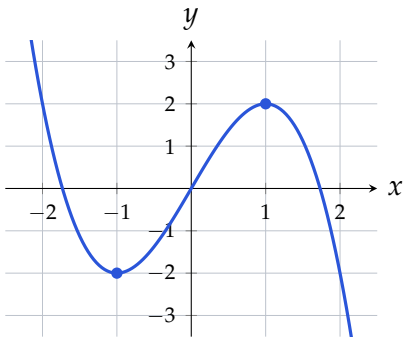


4. Zeros and sign

A **zero** of f is an input a such that $f(a) = 0$. Graphically, the real zeros are the x-intercepts. The function is **positive** where its graph is above the x-axis and **negative** where it is below. A

function whose graph has no breaks can change sign only at a __zero__.

Example 1 • Reading a graph



The graph of $f(x) = -x^3 + 3x$ is shown. Find (a) the intervals where f is increasing and decreasing, (b) where the graph is concave up and concave down, (c) the zeros, (d) where $f(x) > 0$.

(a) Increasing on $(-1, 1)$; decreasing on $(-\infty, -1)$ and $(1, \infty)$.
(b) Concave up on $(-\infty, 0)$; concave down on $(0, \infty)$. The change happens at the point $(0, 0)$.
(c) $-x^3 + 3x = -x(x^2 - 3) = 0$ gives $x = 0$ and $x = \pm\sqrt{3} \approx \pm 1.73$.
(d) $f(x) > 0$ on $(-\infty, -\sqrt{3}) \cup (0, \sqrt{3})$.

Example 2 • Reading a table

x	0	1	2	3	4	5
$g(x)$	-6	-1	2	3	2	-1

Describe how g changes. Where must g have zeros (assume the graph of g has no breaks)?

The outputs increase from $x = 0$ to $x = 3$, then decrease. The changes $5, 3, 1, -1, -3$ decrease steadily, so the graph is concave down.
The sign changes between $x = 1$ and $x = 2$ and between $x = 4$ and $x = 5$: g has a zero in each of these intervals.

Example 3 • In context

The temperature T (°C) of a cup of coffee is $90, 70, 56, 46, 40$ at $t = 0, 5, 10, 15, 20$ minutes. Describe T.

T is decreasing (the coffee cools). The changes are $-20, -14, -10, -6$: the temperature falls more and more slowly, so the rate of change is increasing and the graph is concave up.

Common mistakes

- Writing intervals with y-values instead of x-values.
- Confusing "decreasing" with "concave down". A graph can be increasing and concave down at the same time.

Check your understanding

1. Outputs at $x = 0, 1, 2, 3$ are $10, 7, 5, 4$. The function is __decreasing__ and its graph is concave __up__.
2. The zeros of $f(x) = (x - 2)(x + 5)$ are __$x = 2$ and $x = -5$__.